0.1 The Probabilistic Method

Definition 0.1.1. *A graph $G = (V, E)$ is* sparse *if $|E| \leq 3|V| - 6$.*

Theorem 0.1.1 (Turán)**.** *Every graph on n vertices with more than $\frac{n^2}{4}$ edges contains a triangle.*

Proof. Let G be triangle-free with n vertices and maximum number of edges. For every edge uv, the neighborhoods satisfy $N(u) \cap N(v) = \varnothing$, hence $d(u) + d(v) \leq n$. Summing over all edges gives

$$\sum_{uv \in E} \big(d(u) + d(v)\big) = \sum_{v \in V} d(v)^2 \leq n|E|.$$

By Cauchy–Schwarz, $\sum_v d(v)^2 \geq (2|E|)^2/n$, so $(2|E|)^2 \leq n^2|E|$, which yields $|E| \leq n^2/4$. $\square$ $\square$

Lemma 0.1.1. *If a graph has n vertices and e edges, then its average degree is $2e/n$.*

The celebrated Erdős–Rényi bound shows that the random graph $G(n, p)$ with $p = \frac{\log n}{n}$ is almost surely connected.